Date: 07 July 2026	**Team ID:** SB-7429
Project Name: Smart Lender: Applicant Credibility Prediction for Loan Approval	**Maximum Marks:** 5 Marks

PROJECT PROPOSAL (PROPOSED SOLUTION)

Proposed Solution Proposal Report

Project Overview:

Objective: Build a machine learning-based loan eligibility prediction application (Smart Lender) to evaluate applicant profiles and make data-driven underwriting decisions instantly.

Scope: Data preprocessing (mean/mode imputation), label mapping, multi-model benchmarking (Decision Tree, RF, KNN, XGBoost), feature scaling, serialization of the best model (XGBoost), design of a dark-themed multi-tab Flask dashboard showing evaluation forms and EDA plots.

Problem Statement:

Traditional loan underwriting is manual, slow, and prone to subjective errors, leading to credit bottlenecks and bad debt. A fast, automated, data-driven underwriting tool is needed to filter out default risks.

Resource Requirements Matrix

Resource Type	**Component Description**	**Specification / Allocation**
Hardware	Processor	Intel i3 or above (64-bit OS)
Hardware	System Memory	4 GB RAM minimum (8 GB recommended)
Hardware	Storage	10 GB free space for datasets, virtual environments, and python modules
Software	Operating System	Windows 10/11, macOS, or Linux
Software	Libraries	Python 3.8+, Flask, Pandas, NumPy, Scikit-learn, XGBoost, ReportLab, Seaborn
Data Source	Loan Predictor	Loan Prediction Dataset (train.csv). 614 applicant rows, 13 features.